P-L-00790a (DSA-DM)
January 15, 1971
INTERIM REVISION OF
Fed. Spec. P-L-790
December 13, 1956

INTERIM FEDERAL SPECIFICATION

LUBRICANT, SILICATE CEMENT, DENTAL

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification P-L-790.

1. SCOPE

1.1 Scope. This specification covers a dental lubricant prepared to protect freshly inserted silicate restorations against moisture, and to lubricate strips and disks during finishing.

2. APPLICABLE DOCUMENTS

2.1 Specifications. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing For Drugs, Chemicals, and Pharmaceuticals.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

FSC 6520

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Auburn, Washington, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION

NATIONAL FORMULARY

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America

(Appilcation for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

UNITED STATES DEPARTMENT OF HEALTH, EDUCATION, AND

WELFARE: FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D.C. 20204).

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3. REQUIREMENTS

3.1 Materials. Materials shall be of the highest quality throughout and suitable in all respects for the purpose intended. Any ingredients entering into the lubricant which are not included in the Pharmacopeia of the United States of American (U.S.P.) or National Formulary (N.F.) shall be of quality suitable for pharmaceutical use.

3.1.1 Lubricating compound. The lubricating compound shall be manufactured from either Composition A (see 3.1.1.1) or Composition B (see 3.1.1.2) providing differing ingredients from one formula are not transferred to the other formula.

3.1.1.1 Composition A.

White petrolatum, U.S.P.	.40 percent
Hydrogenated vegetable oil	.50 percent
Theobroma oil, U.S.P.	5 percent
White wax, U.S.P.	5 percent
Flavoring ingredients, U.S.P.	0.5 part (by formula)

The white petrolatum, theobroma oil, white wax and flavoring ingredients shall be in accordance with the tests, standards, and requirements of the U.S.P. including any supplements or revisions thereto.

3.1.1.2 Composition B.

White petrolatum, U.S.P.	.40 percent
Hydrogenated vegetable oil	.41.5 percent
Silicone fluid	.10.0 percent
Semi-bleached beeswax	.7.5 percent
Flavoring ingredients	1.0 percent

The white petrolatum shall be in accordance with the tests, standards, and requirements of the U.S.P. including any supplements or revisions thereto. The silicone fluid shall conform to the Food and Drug Administration, Federal Food, Drug, and Cosmetic Act for applications indicated in the food additive regulations (see 6.2).

3.2 Description. Shall be a dental lubricating compound suitable for use to protect freshly inserted silicate restorations against moisture, and to lubricate strips and disks during finishing procedures. The compound shall be of such consistency that, when applied to the teeth in the usual manner, it will not liquefy and run on the teeth.

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3.2.1 Immediate container. Immediate container shall be collapsible metal tube conforming to group B, class 1, type a, closure A or D of PPP-C-186a except the cap liner will not be required when the neck of the tube and inner face of cap are so designed as to form an effective seal which will permit reclosure and prevent loss of contents. Tube shall contain a minimum of 1 ounce of lubricant.

3.3 Physical properties.

3.3.1 Extrusion after softening. The lubricating compound shall not liquefy when tested as specified in 4.4.1.

3.3.2 Extrusion after chilling. Excessive pressure shall not be required to extrude material from tube after testing as specified in 4.4.2.

3.3.3 Freezing and thawing effects. The material shall show no adverse effects such as graininess, separation, or loss of smooth texture when tested as specified in 4.4.3.

3.4 Delivery schedule. No more than 2 months shall have elapsed from date of manufacture of the lubricant to date of delivery to the Government.

3.5 Lot or batch number. Each lubricant tube shall be marked with the lot or batch number.

3.6 Stability. The lubricant shall comply with all the requirements of this specification for no less than two years when stored in a cool, dry place (50 to 80 degrees Fahrenheit).

3.7 Irritation. Unless otherwise specified (see 6.1), prior to delivery the manufacturer shall furnish a statement directly to the Technical Services Branch, (Code ATTS-2), Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania, 19101, stating that the material does not normally irritate oral tissues and does not contain poisonous ingredients in sufficient concentration to be harmful to human beings when used as directed.

3.8 Workmanship. Workmanship shall be first class throughout. The lubricant shall be smooth and homogeneous and shall be free from defects which may detract from its appearance or may impair its serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials and packaging materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Inspection of raw materials. In accordance with 4.1, the supplier is responsible for insuring that raw materials used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.1.5 Manufacturer's records. The percentage of the raw materials listed in 3.1.1.1 and 3.1.1.2 shall be ascertained from the manufacturer's batch formula control records.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I and III (see 4.2.1 and 4.3) and table II (see 4.2.2).

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4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. Unit of product for examination purposes shall be unit of issue (tube of lubricant).

Table I. Inspection and Acceptable Quality Levels

Type defect	Inspection level	AQL (Percent defective)
Major	S-2	2.5
Minor	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be unit of issue (one tube of lubricant).

Table II. Tests

Characteristic	Reqmt.	Test procedure	Inspection level
Extrusion after softening	3.3.1	4.4.1	S-1
Extrusion after chilling	3.3.2	4.4.2	S-1
Freezing and thawing effects	3.3.3	4.4.3	S-1

4.3 Examination. The tube of lubricant shall be examined for defects, including, but not limited to, those listed in table III.

Table III. Classification of Defects

Categories	Defects
Major:	
101	Amount of contents in immediate container not as specified.
102	Splits, rips, tears, or cracks present in immediate container.
103	Cap does not fit tightly over neck of tube.
104	Closure not of type specified.
105	Objectionable foreign material present in lubricant.
Minor:	
201	Packaging and packing not as specified.
202	Marking not as specified.

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4.4 Tests.

4.4.1 Extrusion after softening. A tube of the paste shall be submerged in a water bath at 37 ± 0.1 °C. for 30 minutes. The tube shall then be removed from the bath and a portion (no less than 1/2 inch in length) shall be extruded onto a glass plate measuring approximately 4 inches by 5 inches.

4.4.2 Extrusion after chilling. A tube of paste shall be submerged in a water bath at 15 ± 0.2 °C. for 30 minutes. The tube shall then be removed from the bath and a portion extruded as described in 4.4.1.

4.4.3 Freezing and thawing effects. A tube of paste shall be refrigerated at -4 ± 0.2 °C. for 24 hours. The tube shall then be allowed to thaw for 12 hours.

5. PREPARATION FOR DELIVERY

5.1 Preparation For Delivery shall be in accordance with PPP-C-186a. Procedure code No. 14 applies.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) If statement of irritation is to be furnished to another Government activity (see 3.7).
(c) Selection of applicable levels of packaging and packing (see 5.1).

6.2 A silicone fluid suitable for use in the lubricant is General Electric Co. SF-18 silicone fluid.

6.3 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.4 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Federal Government requirements.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specifica tion in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION
P-L-00790a(DSA-DR) FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

DD FORM 1 JAN 84 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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